

Trainings ▾

General Information

The QSK45, QSK60, and QSK60 two-stage exhaust manifolds have been aerodynamically designed to reduce pumping losses and enhance pulse flow. In addition, the manifold connection has been improved to allow for thermal expansion. Stainless steel bellows connections have been added to numerous locations in the QSK45, QSK60, and QSK60 two-stage exhaust systems. These premium connections will minimize the possibility of a cracked exhaust manifold due to thermal expansion.

Metric drilled and tapped bosses are available in the cylinder head exhaust port to measure exhaust temperatures that will be monitored by the CENSE™ engine monitoring system.

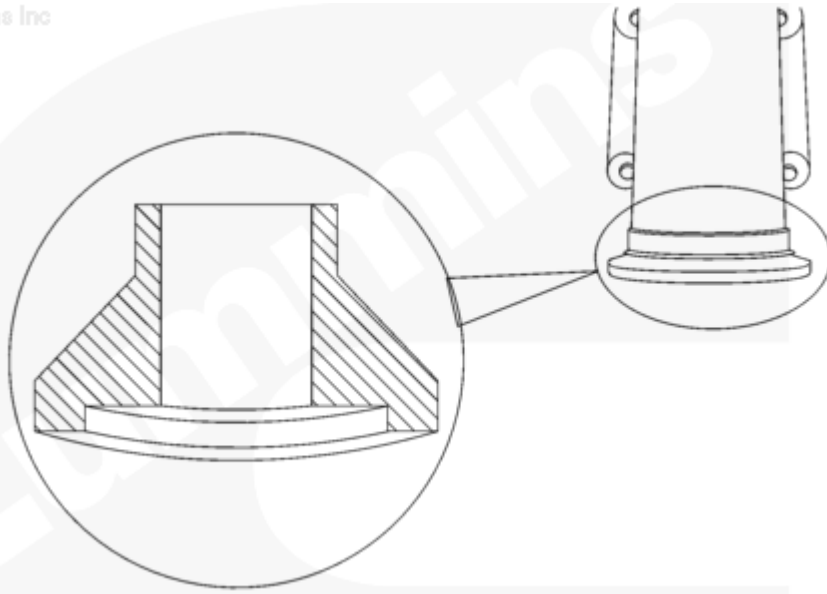
The QSK45 exhaust outlet connections are 139.5 mm [5.5 in] in diameter.

The QSK60 single stage has two 226 mm [8.9 in] inside diameter outlets for either top-out or side-out orientations. The exhaust outlet connection flange has eight 18 mm [0.71 in] diameter holes on a 284 mm [11 in] diameter bolt circle.

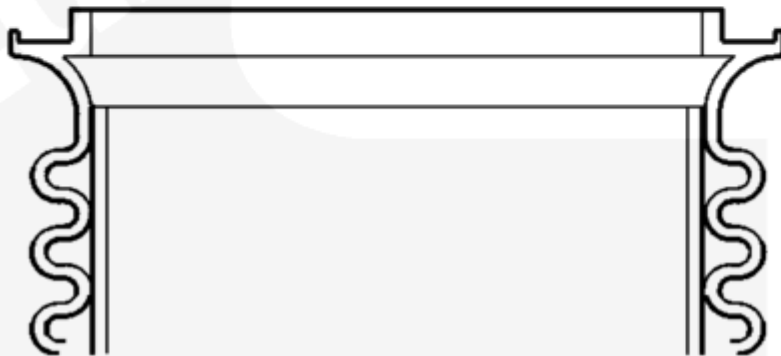
The QSK60 two stage uses a 45-degree exhaust outlet connection. The exhaust outlet connection flange has twelve 19 mm [0.75 in] diameter holes on a 362.0 mm [14.25 in] diameter bolt circle.

Marine Applications

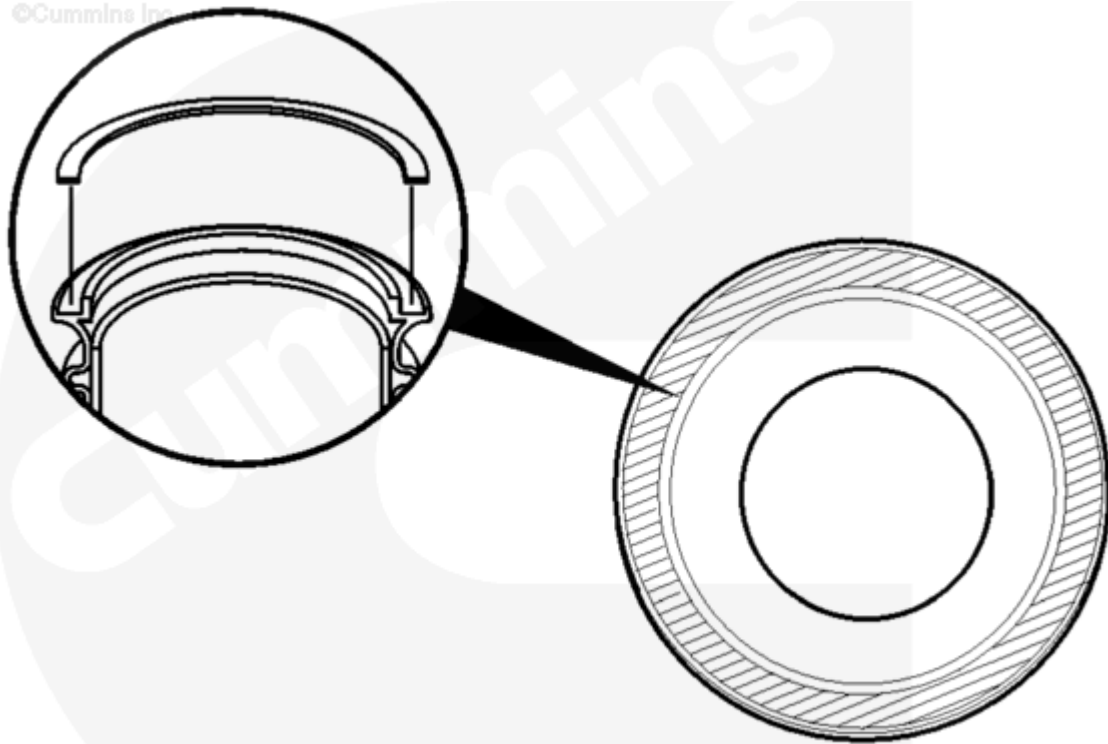
The QSK60M engine uses a dry, spigot, and face seal design exhaust manifold similar to the K38 and K50 marine engines. The bellows in this design use a gasket between it and the manifold sections. Reference QuickServe® Online for the proper gasket part numbers. This design is **only** used on marine engines and is **not** interchangeable with standard industrial and power generation exhaust designs. See the illustrations below.



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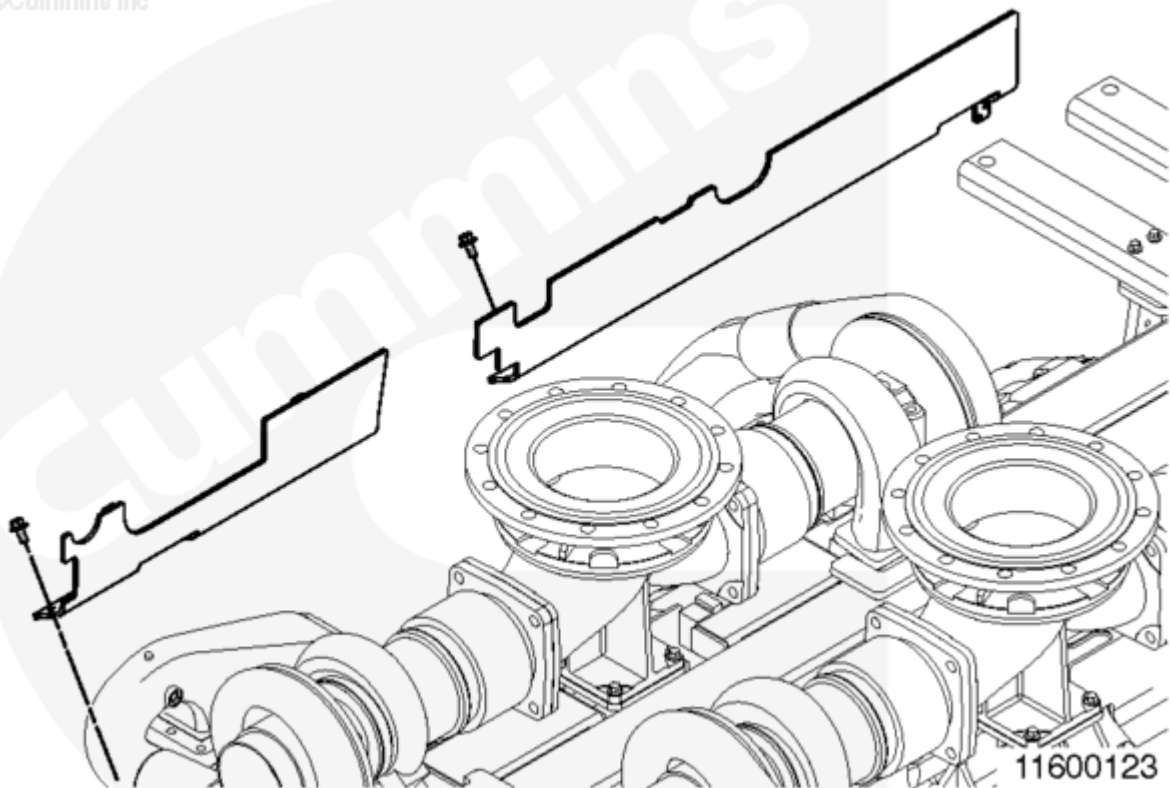
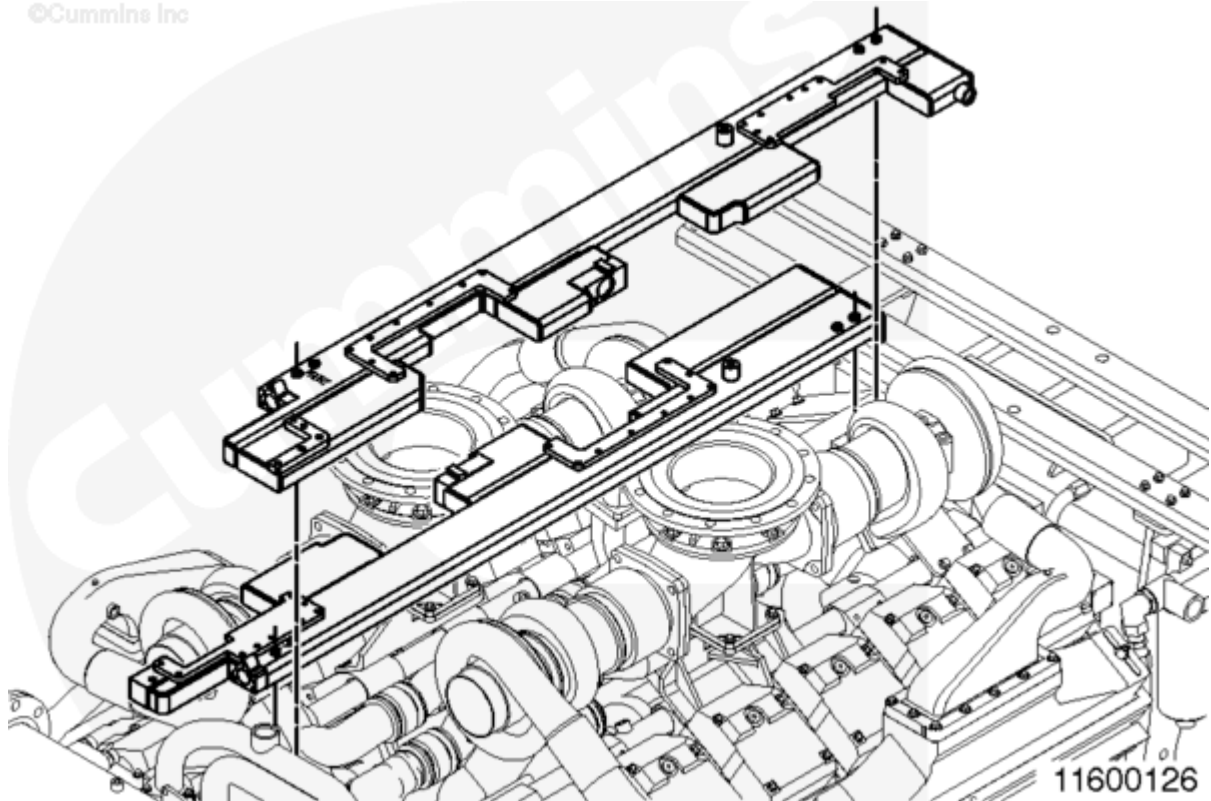
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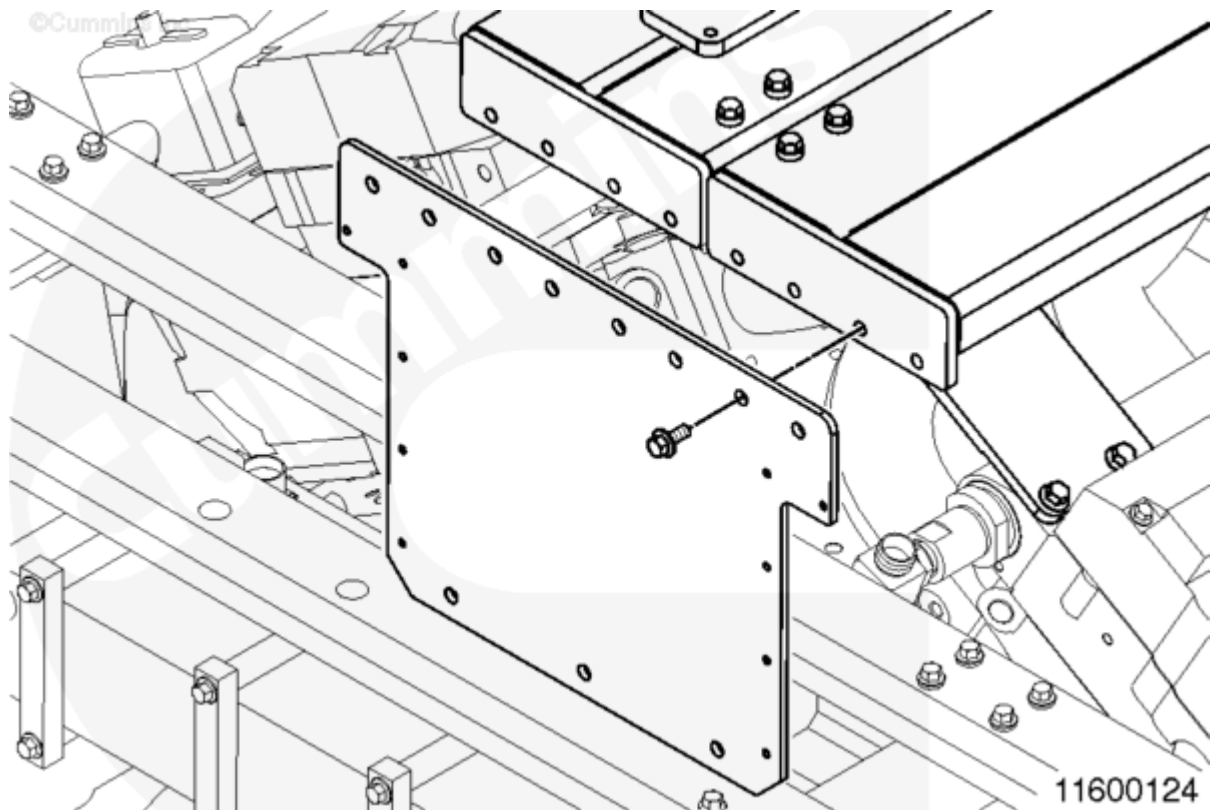


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The exhaust manifolds are shielded with LTA coolant-cooled heat shields to meet Marine Agency requirements. There are two shields that extend the length of the engine. There are also a number of other dry heat shields mounted behind the cylinder heads and at the end of the vee cavity above the flywheel housing.

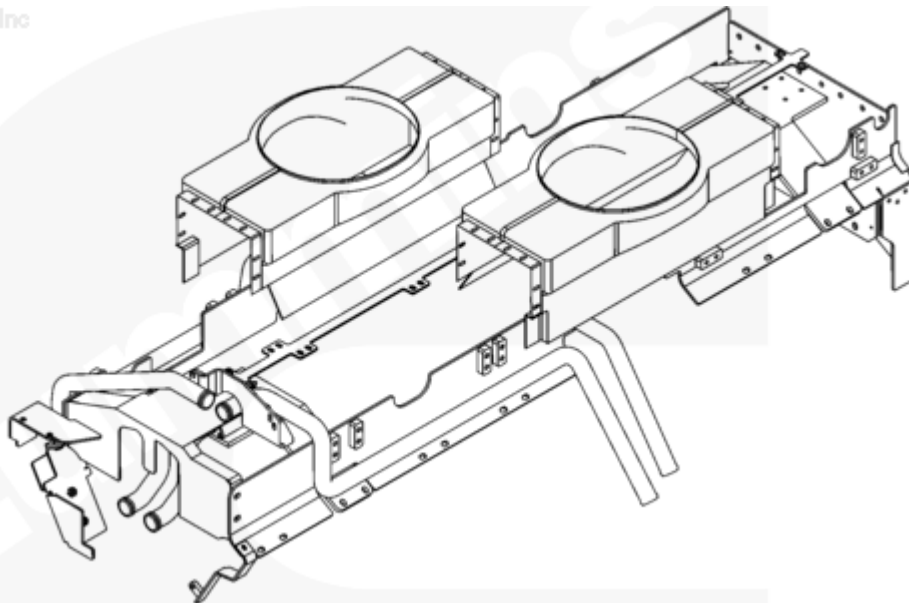
The following three illustrations show the original exhaust manifold heat shield design utilized on the mechanically actuated injector marine application engine.



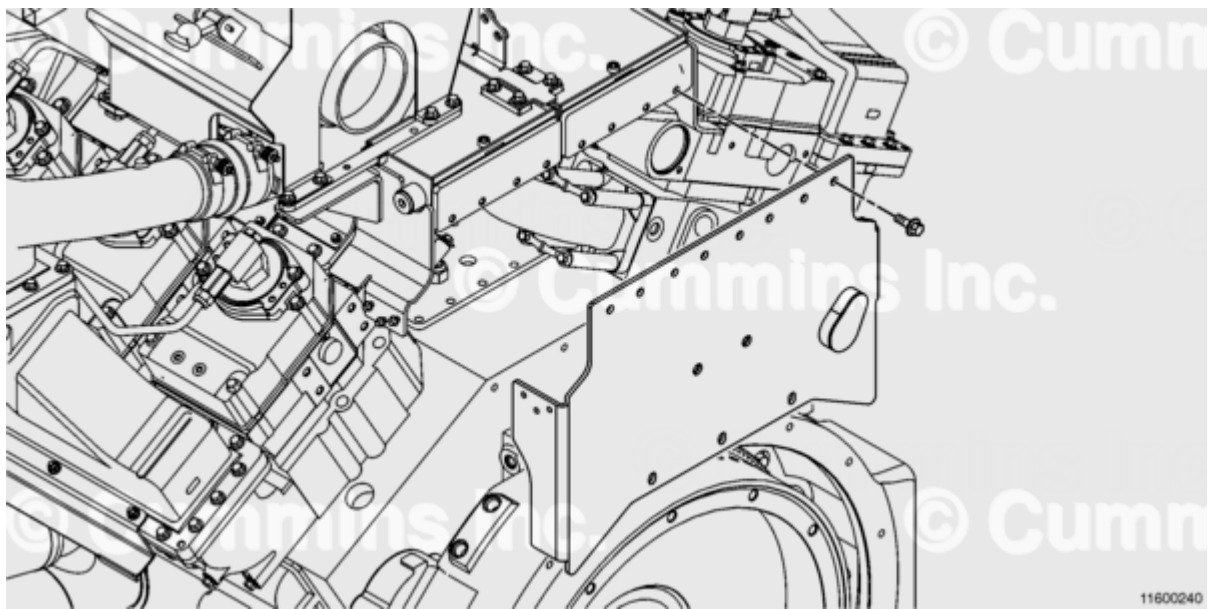
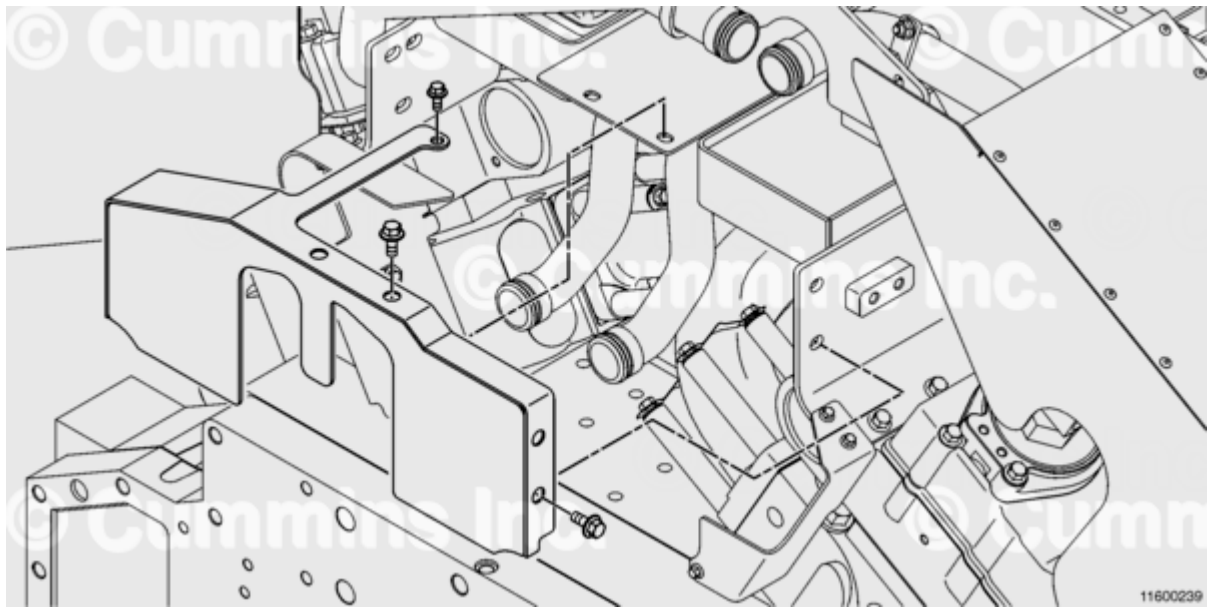


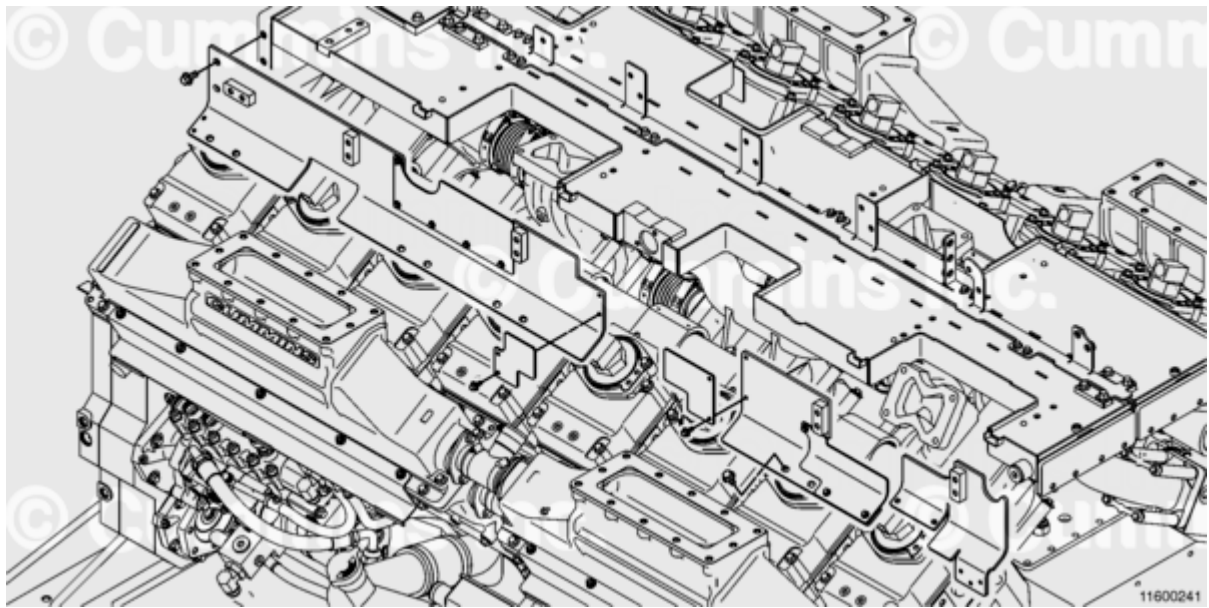
The exhaust manifold heat shields have been redesigned to allow for optimized plumbing arrangements and shielding capability. The illustration below illustrates the present arrangement utilized on the mechanically actuated injector marine application engine.

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The exhaust manifold heat shields have undergone further revisions for integration into the electronically actuated injector engines. These redesigns included provisions to allow for improved access to the turbocharger mounting flanges and the exhaust shield coolant connections to the thermostat housing, as well as to improve upon the shielding capabilities of the present design. The illustrations below illustrate some of these improvements.





Last Modified: 18-Nov-2008
